

Internal Model Based PID Control of Shell and Tube Heat Exchanger System

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Abstract— Shell and tube heat exchanger system is widely used in chemical plants because it can sustain wide range of temperature and pressure. The main purpose of a heat exchanger system is to transfer heat from a hot fluid to a cooler fluid, so temperature control of outlet fluid is of prime importance. To control the temperature of outlet fluid of the shell and tube heat exchanger system a conventional PID controller can be used, for better disturbance rejection and more optimal control a feed forward controller can also be employed. Due to inherent disadvantages of conventional control techniques, model based control technique is employed and an internal model based PID controller is developed to control the temperature of outlet fluid of the shell and tube heat exchanger system. The designed controller regulates the temperature of the outgoing fluid to a desired set point in the shortest possible time irrespective of load and process disturbances, equipment saturation and nonlinearity. The internal model based PID controller provides a satisfactory performance in both steady state and transient state. The developed internal model based PID controller has demonstrated 84% improvement in the overshoot and 44.6% improvement in settling time as compared to the classical controller.

Keywords- Internal model based PID control, PID controller, Shell and tube heat exchanger system

I. INTRODUCTION

In practice, all chemical processes involve production or absorption of energy in the form of heat. Heat exchanger is commonly used in chemical processes to transfer heat from a hot fluid through a solid wall to a cooler fluid. There are different types of heat exchanger used in the industry but most of the industry use shell and tube type heat exchanger system [5].

Shell-and-tube heat exchangers are probably the most common type of heat exchangers applicable for a wide range of operating temperatures and pressures. They have larger ratios of heat transfer surface to volume than double-pipe heat exchangers, and they are easy to manufacture in a large variety of sizes and configurations. They can operate at high pressures, and their construction facilitates disassembly for periodic maintenance and cleaning. Shell-and-tube heat exchangers find widespread use in refrigeration, power generation, heating and air conditioning, chemical processes, manufacturing, and medical applications. A shell-and-tube heat exchanger is an extension of the double-pipe configuration. Instead of a single pipe within a larger pipe, a

shell-and-tube heat exchanger consists of a bundle of pipes or tubes enclosed within a cylindrical shell. In shell and tube heat exchanger one fluid flows through the tubes, and a second fluid flows within the space between the tubes and the shell [6].

This paper reports a work that considers a shell and tube heat exchanger and builds a single input-single output model of the system with the help of experimental data. The outlet temperature of the shell and tube heat exchanger system has to be kept at a desired set point according to the process requirement. Firstly, a classical PID controller is implemented in a feedback control loop so as to achieve the control objectives. PID controller exhibits high overshoots which is undesirable. To reduce the overshoot feed forward controller is used along with the feedback controller. The combined effect of feedback and feed forward control scheme gives a much better result than the feedback PID controller. Still the overshoot remains in a higher side. To further minimize the overshoot internal model based controller and internal model based PID controller is implemented. In the model based controller the process model is implemented in parallel with the real process. The internal model controller based controller design has gained widespread acceptance because it has only a single tuning parameter namely the closed loop time constant λ . The controller is designed according to a model of the actual process. The internal model controller reduces the overshoot and settling time. In this research paper four type of controller are designed to achieve the control objective and a comparative study between the controllers are evaluated.

II. SHELL AND TUBE HEAT EXCHANGER SYSTEM

A typical interacting chemical process for heating consists of a chemical reactor and a shell and tube heat exchanger system. The process fluid which is the output of the chemical reactor is stored in the storage tank. The storage tank supplies the fluid to the shell and tube heat exchanger system using a pump and a non returning valve. The heat exchanger heats up the fluid to a desired set point using super heated steam at 180°C supplied from the boiler. The super heated steam comes from the boiler and flows through the tubes, whereas, the process fluid flows through the shells of the shell and tube heat exchanger system.

Different assumptions have been considered in this research paper. The first assumption is that the inflow and the

outflow rate of fluid are same, so that the fluid level is maintained constant in the heat exchanger. The second assumption is the heat storage capacity of the insulating wall is negligible. In this feedback process control loop, the controller is reverse acting, the valve used is of air to open (fail-close) type. A thermocouple is used as the sensing element, which is implemented in the feedback path of the control architecture. The temperature of the outgoing fluid is measured by the thermocouple and the output of the thermocouple (voltage) is sent to the transmitter unit, which eventually converts the thermocouple output to a standardized signal in the range of 4-20 mA. This output of the transmitter unit is given to the controller unit. The controller implements the control algorithm, compares the output with the set point and then gives necessary command to the final control element via the actuator unit. The actuator unit is a current to pressure converter and the final control unit is an air to open (fail-close) valve. The actuator unit takes the controller output in the range of 4-20 mA and converts it in to a standardized pressure signal, i.e in the range of 3-15 psig. The valve actuates according to the controller decisions. Figure 1 shows the basic feedback control scheme implemented in a shell and tube heat exchanger system.

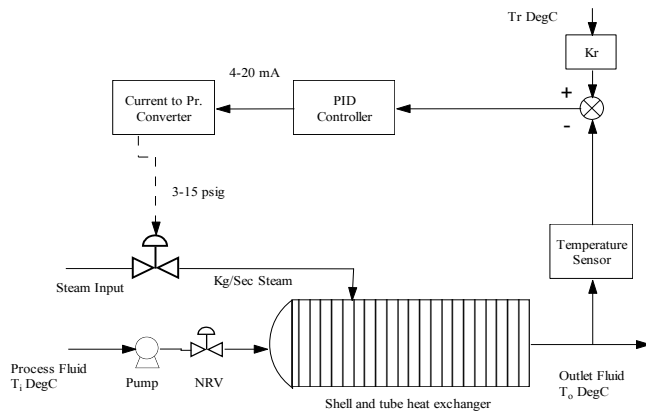


Figure 1: Shell and tube heat exchanger system control scheme

There can be two types of disturbances in this process, one is the flow variation of input fluid and the second is the temperature variation of input fluid. But in practice the flow variation of input fluid is a more prominent disturbance than the temperature variation in input fluid. So, in feed forward control loop, the input fluid flow is measured and the disturbance in the flow is controlled using a feed forward controller. The output of the feedback and the feed forward controller is added and the resultant output is given to the control valve.

III. MATHEMETICAL MODELING

In this section, the heat exchanger system, actuator, valve, sensor are mathematically modelled using the available experimental data. The experimental process data is summarized below [10].

Exchanger response to the steam flow gain	50°C/(kg/sec)
Time constants	30 sec

Exchanger response to variation of process fluid flow gain	1°C/(kg/sec)
Exchanger response to variation of process temperature gain	3°C/°C
Control valve capacity	1.6 kg/sec for steam
Time constant of control valve	3 sec
The range of temperature sensor	50°C to 150°C
Time constant of temperature sensor	10 sec
From the experimental data, the transfer functions and gains are obtained as below.	
Transfer function of process	$\frac{50e^{-s}}{30s+1}$
Gain of valve	0.13
Transfer function of valve	$\frac{0.13}{3s+1}$
Gain of current to pressure converter	0.75
Transfer function of disturbance variables (flow and temperature disturbances respectively)	$\frac{1}{30s+1}, \frac{3}{30s+1}$
Transfer function of thermocouple	$\frac{0.16}{10s+1}$

To control the exit temperature of the heat exchanger system a classical feedback controller is used. The PID control algorithm remains the most popular approach for industrial process control despite continual advances in control theory. This is not only due to the simple structure which is conceptually easy to understand and, which makes manual tuning possible, but also to the fact that the algorithm provides adequate performance in the vast majority of applications. Figure 2 represents the transfer function block diagram of feedback control of shell and tube heat exchanger system.

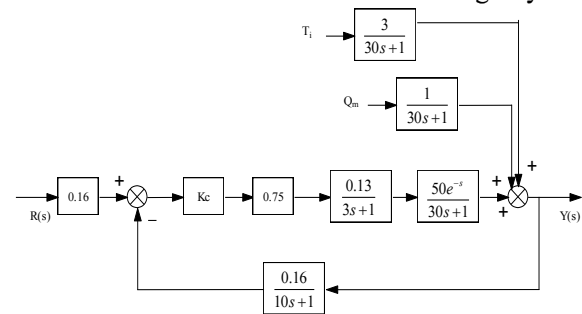


Figure 2: Feedback control of shell and tube heat exchanger system
The characteristic equation $1+G(s)H(s)=0$ in this case is obtained as below.

$$900s^3 + 420s^2 + 43s + 0.798K_{cu} + 1 = 0 \quad (1)$$

Routh stability criterion gives K_{cu} as 23.8.

$$\text{Auxiliary equation } 420s^2 + 0.798K_{cu} + 1 = 0 \quad (2)$$

$$\omega = 0.218 \text{ and } T = 28.79$$

PID controller in continuous time is given as

$$u(t) = K_c \left(e(t) + \frac{1}{\tau_i} \int_0^t e(t) dt + \tau_d \frac{de(t)}{dt} \right) \quad (3)$$

Most of the PID tuning rules developed in the last sixty years use frequency-response methods [2,4]. Examples

include, Ziegler–Nichols rule, symmetric optimum rule, Ziegler–Nichols' complementary rule, some-overshoot rule, no-overshoot rule, refined Ziegler–Nichols rule, integral of squared time weighted error rule, and integral of time absolute error rule [3]. These methods are straightforward to apply since they provide simple tuning formulae to determine the PID controller parameters. According to Zeigler-Nichols frequency response tuning criteria $K_p = 0.6K_{cu}$, $\tau_i = 0.5T$ and $\tau_d = 0.125T$. For the PID controller in the heat exchanger, the values of tuning parameters obtained are $K_p=14.28$, $\tau_i=14.395$, $\tau_d=3.59$. Usually, initial design values of PID controller obtained by all means needs to be adjusted repeatedly through computer simulations until the closed loop system performs or compromises as desired. In order to have a good closed-loop time response, the following performance functions are considered during the design of a PID controller.

$$J(K_p, K_i, K_d) = \int_0^{\infty} (r(t) - y(t))^2 dt \quad (4)$$

$$J(K_p, K_i, K_d) = \int_0^{\infty} t |(r(t) - y(t))| dt \quad (5)$$

Thus, the optimal PID controller design is stated as $\min J(K_p, K_i, K_d)$ (6)

The disturbance input introduces error in the system performance. In several systems the disturbance can be predicted and its effect can be eliminated with the help of feed forward controller before it can change the output of the system. In feedback control scheme the sensor detects the process output and gives the error to the controller, which in turn takes appropriate controlling action. But till the controlling action reaches the process, the output has been changed. So a feed forward controller along with the feedback controller has been implemented. A feed forward control estimates the error and changes the manipulating variable before the disturbance can affect the output. To further minimize the overshoot a feed-forward controller is introduced in the forward path of the process along with the feedback controller. The combined effect of feedback and feed-forward controller reduces the overshoot value. Figure 3 shows the transfer function representation of system with feedback and feed-forward controller.

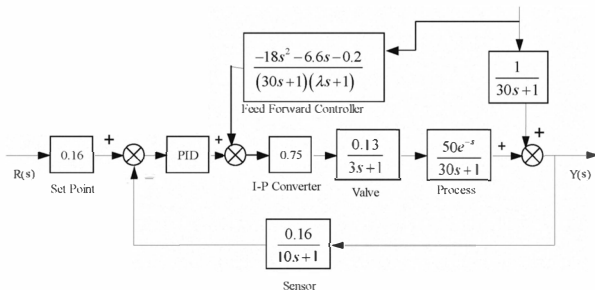


Figure 3: Feed-forward plus feedback control block diagram of shell and tube heat exchanger system

$$G_p(s) = \frac{5e^{-s}}{90s^2 + 33s + 1}, \quad G_d(s) = \frac{1}{30s + 1}$$

The transfer function of the feed-forward controller is

$$G_{cf}(s) = \frac{-G_d(s)}{G_p(s)} \quad (7)$$

$$G_{cf}(s) = \frac{-18s^2 - 6.6s - 0.2}{(30s + 1)(\lambda s + 1)} \quad (8)$$

Here, 'λ' is filter parameter. Its range is from 0 to 1. It has been used to make the transfer function semi proper.

IV. INTERNAL MODEL CONTROLLER

Internal model controller provides a transparent framework for control system design and tuning. The structure of internal model controller is shown in figure 4. The main feature of internal model controller is that the process model is in parallel with the actual process. The transfer function of the process is shown in eq (9).

$$G_p(s) = \frac{5e^{-s}}{90s^2 + 33s + 1} \quad (9)$$

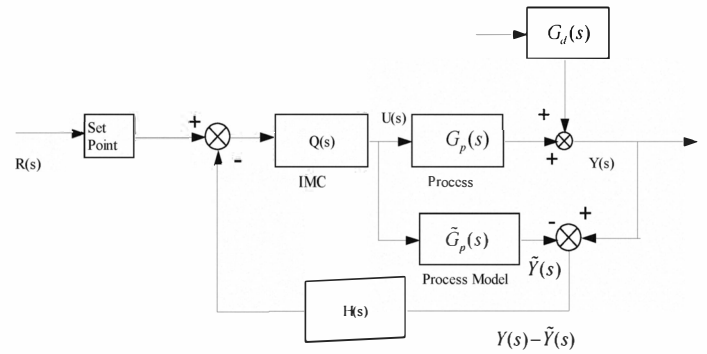


Figure 4: Internal model control

The process model $\tilde{G}_p(s)$ is factored in to two parts, that is invertible part $\tilde{G}_{p+}(s)$ and non invertible part $\tilde{G}_{p-}(s)$. The non invertible part consists of RHP zeros and time delays. This factorization is performed so as to make the resulting internal model controller stable.

$$\tilde{G}_p(s) = \tilde{G}_{p-}(s)\tilde{G}_{p+}(s) \quad (11)$$

$$\tilde{G}_p(s) = \frac{5}{(30s + 1)(3s + 1)(0.083s^2 + 0.5s + 1)} (-0.083s^2 - 0.5s + 1) \quad (12)$$

$$Q(s) = \tilde{G}_{p-}^{-1}(s)f(s) \quad (13)$$

$$Q(s) = \frac{(30s + 1)(3s + 1)(0.083s^2 + 0.5s + 1)}{5} \frac{1}{(\lambda s + 1)^4} \quad (14)$$

In practice λ is taken as one third of one fifth of the time constant [9]. So, the values of λ are obtained as 11.4 and 17. Substituting the values of λ as 11.4 and 17 in eq(14) we get

$$Q(s) = \frac{1.494s^4 + 603s^3 + 235.966s^2 + 6.7s + 0.2}{16889.6s^4 + 5926.17s^3 + 779.76s^2 + 45.6s + 1} \quad (15)$$

$$Q(s) = \frac{1.494s^4 + 603s^3 + 235.966s^2 + 6.7s + 0.2}{83521s^4 + 19652s^3 + 1734s^2 + 68s + 1} \quad (16)$$

The transfer function of internal model controller denoted by Q(s) for different values of the filter parameter is shown in eq (15) and eq (16).

V. INTERNAL MODEL BASED PID CONTROLLER

The most common industrial controller is PID controller and internal model control scheme shown in figure 4 can be used to design a standard feedback PID controller. The standard feedback controller is a function of the internal model $\tilde{G}_p(s)$ and internal model controller $Q(s)$. The transfer function of feedback controller equivalent to internal model controller is shown in eq (17).

$$G_c(s) = \frac{Q(s)}{1 - \tilde{G}_p(s)Q(s)} \quad (17)$$

Substituting the values in eq (17) the transfer function of the feedback controller based on internal model controller is obtained as shown in eq (18).

$$G_c(s) = \frac{90s^2 + 33s + 1}{5(\lambda^2 s^2 + 2\lambda s)} \quad (18)$$

Equating the terms of eq (3) and eq (18), the values of PID parameters are obtained. PID controller for improved disturbance rejection can be achieved by changing the internal model controller filter. For improved disturbance rejection the filter is in the form of $f(s) = \frac{\gamma s + 1}{(\lambda s + 1)^n}$ (19)

γ is selected to achieve good disturbance rejection. It is selected to cancel slow disturbance time constant. Several methods of tuning of internal model controller and efficient calculation of filter parameter have been proposed in literature [1,7,8,9].

VI. SIMULATIONS AND TESTING

The simulations for the different control mechanism discussed above were carried out in Simulink and the simulation results have been obtained. Figure 5 represents the Simulink modelling of the shell and tube heat exchanger system with PID as a feedback controller.

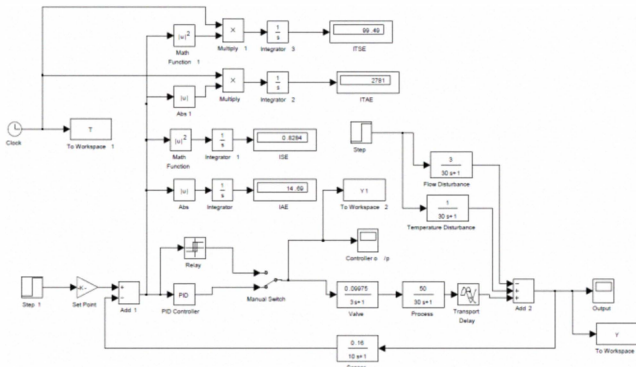


Figure 5: Simulink model of shell and tube heat exchanger system with feedback PID controller

. As shown in figure 5 for auto-tuning of PID controller, a relay block is connected in parallel with the PID controller with the help of a manual switch. When the auto-tune function is required, the manual switch is set to the relay block. The behaviour obtained from auto tuning mode is very similar to the behaviour obtained from Zeigler- Nichols closed loop cycling method. Figure 6 (a) shows the unit step response of

the shell and tube heat exchanger system with PID controller with and without transportation delay. Figure 6(b) shows the response of the controller and shell and tube heat exchanger system when auto-tune mode is set.

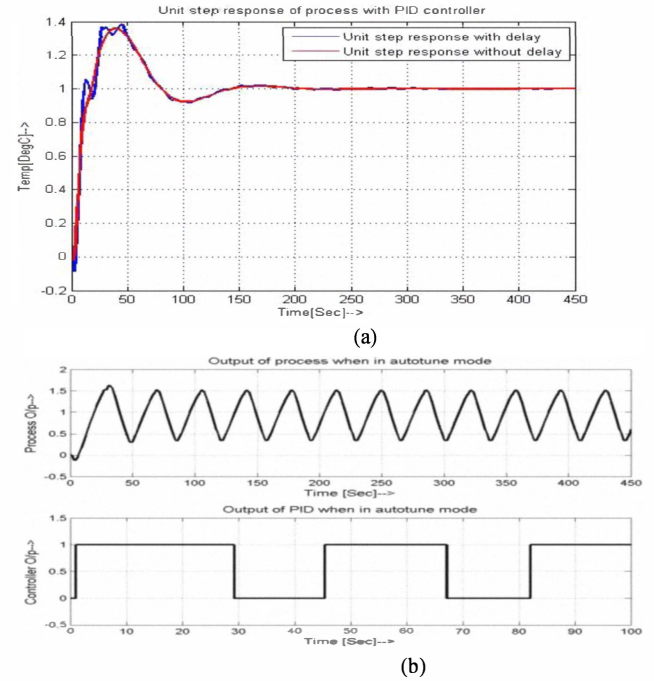


Figure 6: (a) Unit step response of shell and tube heat exchanger system with feedback controller (b) Response of controller and shell and tube heat exchanger system in auto-tune mode

Figure 7 shows the Simulink model of shell and tube heat exchanger system with both feedback and feed-forward controller.

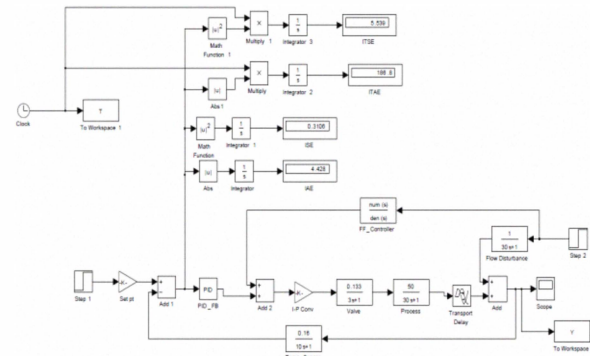


Figure 7: Simulink model of shell and tube heat exchanger system with feedback and feed-forward controller

Figure 8 shows the unit step response of the shell and tube heat exchanger system with feedback and feed forward controller. The combined effect of feedback and feed forward controller reduces the overshoot and also decreases the settling time by 21%.

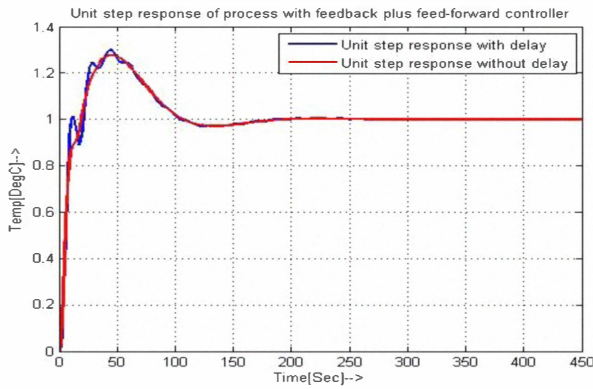


Figure 8: Unit step response of shell and tube heat exchanger system with feedback and feed-forward controller

Figure 9 shows the Simulink modelling of shell and tube heat exchanger system with internal model controller.

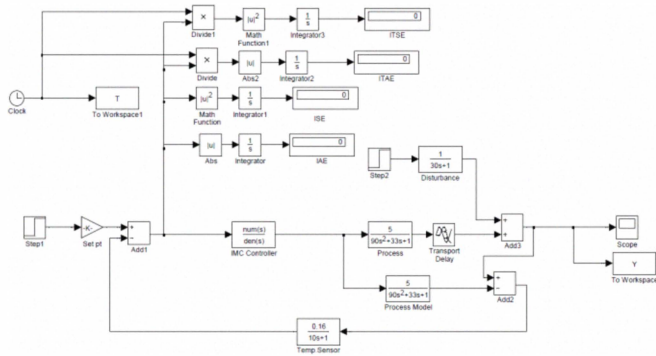


Figure 9: Simulink model of internal model controller

Figure 10 shows the unit step response of the shell and tube heat exchanger system when an internal model controller is implemented in the series of the real process and an approximate model of the process is placed in parallel of the real process. The step response shows less overshoot from feedback controller and feedback plus feed forward controller but it consists of steady state error.

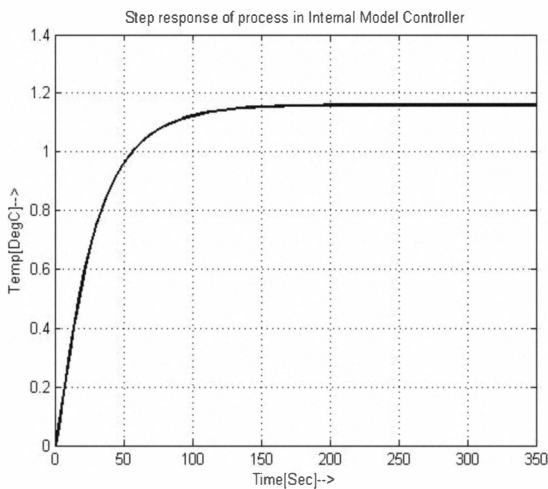


Figure 10: Unit step response of process with internal model controller

Figure 11 shows the Simulink model of shell and tube heat exchanger system with internal model based PID controller. Internal model based PID controller carries a significant advantage over conventional PID controller as there is only one tuning parameter where as there are three tuning parameter in conventional PID controller.

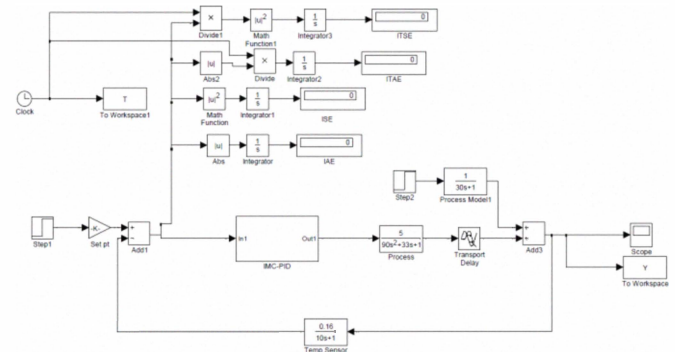


Figure 11: Simulink model of internal model based PID controller

Figure 12 shows the unit step response of shell and tube heat exchanger system when an internal model based PID controller is used to control the controlling variable. The maximum overshoot is only 5% and settling time is also reduced.

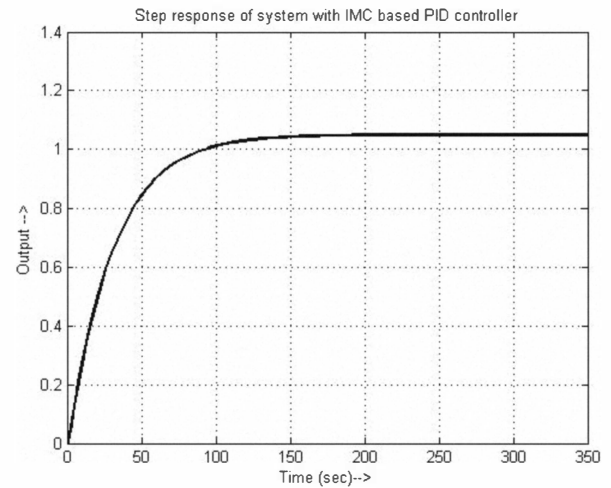


Figure 12: Unit step response of internal model based PID controller

VII. RESULTS AND DISCUSSIONS

To evaluate the performance of the different controllers this paper has considered two vital parameters of the step response of the system. The first parameter is the maximum overshoot and the second parameter is the settling time. A comparative study of their performance has been shown in the table I below.

TABLE I
COMPARISON OF DIFFERENT PARAMETERS IN CONTROLLERS

S.No		Overshoot (%)	Settling Time (sec)
1	Feedback PID	38.38	115.2
2	Feedback plus feed-forward controller	30.04	91.3
3	Internal model controller	16	77.9
4	Internal model based PID controller	5	63.8

Table II gives a comparative study of different performance indices of the four controllers.

TABLE II
COMPARISON OF ERROR CRITERION VALUES OF CONTROL STRATEGIES

S.No		IAE	ISE	ITAE	ITSE
1	Feedback PID	4.755	0.366	192.6	6.333
2	Feedback plus feed-forward controller	4.428	0.31	186.8	5.539
3	Internal model controller	5.86	8.94	200.4	8.37
4	Internal model based PID controller	5.4	8.17	211.6	9.45

Table I shows that feedback PID controller gives very high overshoot of 38%. To compensate this kind of high overshoot a feed forward controller in conjunction with the conventional PID in feedback loop is implemented. By implementing this method the system overshoot was reduced to 30%, an improvement of 21%. Though the overshoot has some what decreased, it can be further decreased by implementing internal model based controller and internal model based PID controller. By implementing internal model based controller and internal model based PID controller the overshoot reduces to 16% and 5% respectively. In feedback controller the settling time was 115.2 sec where as in feed forward plus feedback controller the settling time decreases to 91.34 sec, an improvement of 20.7%. By implementing internal model based controller and internal model based PID controller the settling time decreases by 77.9 sec and 63.8 sec.

VIII. CONCLUSIONS

This paper takes a case study of shell and tube heat exchanger system and evaluates different methods to control the outlet fluid temperature. Four different kind of controllers are designed to control the outlet temperature of fluid and the performances of these controllers are evaluated by two different methods. One of the methods for performance evaluation is the time domain analysis of overshoot and settling time and other method is calculation of performance indices like IAE, ISE, ITAE and ITSE. This paper takes the process model to be the same as the process, which is practically impossible to achieve. So as a further work we can implement direct model and inverse model based controller and apply system identification as well as neural network concepts for estimation of process model.

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